

Investigating Eye-Body Coupling and Gaze Adaptation in Virtual Reality

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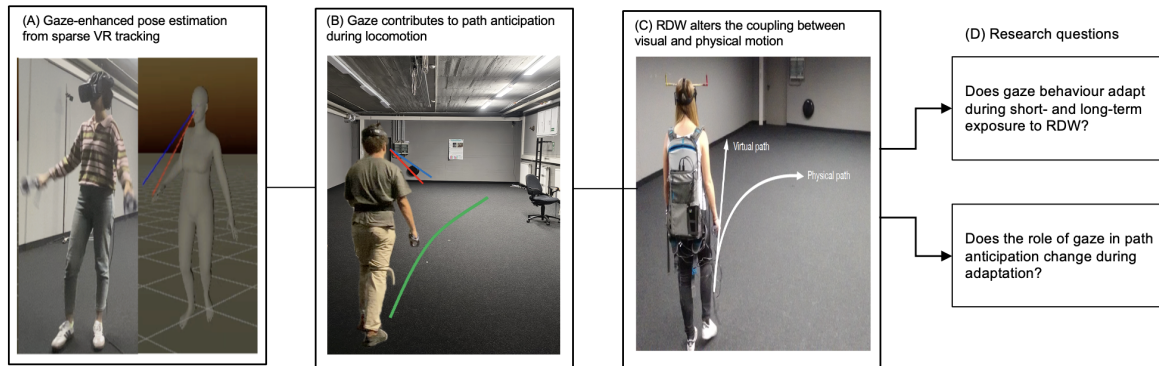


Figure 1: Conceptual overview of the research. (A) Gaze enhances 3D pose estimation from sparse VR tracking [2]. (B) Gaze supports anticipatory cues for walking direction [8, 9]. (C) Redirected walking (RDW) changes the mapping between virtual and physical motion [6, 1]. Building on these foundations, our research examines how gaze behaviour adapts over short- and long-term exposure to RDW, and whether the role of gaze in indicating walking direction changes during adaptation.

ABSTRACT

Gaze plays a central role in guiding human movement, yet most VR locomotion systems still treat gaze and motion as separate signals. Our research examines how these modalities interact during natural and redirected walking, and how this coupling changes under altered visual-motor contingencies. We first showed that integrating binocular gaze with sparse VR tracking enhances full-body pose estimation. Building on this foundation, we now analyze high-frequency gaze behaviour during both short- and long-term exposure to redirected walking (RDW) to investigate whether gaze strategies adapt over time. The overall goal of our research is to characterize how gaze-locomotion coupling reorganizes under RDW and to determine whether gaze remains informative for understanding walking direction throughout adaptation. This paper summarizes our completed and ongoing work and outlines key questions for discussion during the Doctoral Consortium.

Index Terms: Virtual reality, Eye tracking, Redirected walking, Locomotion, Pose estimation, Behaviour prediction

1 INTRODUCTION AND RELATED WORK

Natural locomotion in virtual reality relies on accurate prediction of how users move through space. Redirected walking (RDW) techniques exploit these predictions to subtly guide users away from physical boundaries while preserving the sensation of walking freely [14, 17, 18, 13]. At the same time, decades of visuo-motor research show that gaze systematically leads motor action: people look where they intend to move before the movement unfolds [12, 9]. In VR, this has been confirmed through studies showing that gaze can support steering decisions and short-horizon path prediction [8, 20]. Despite these advances, most VR locomotion

models treat gaze as a static cue—an instantaneous indicator of upcoming motion—rather than as a behaviour that may itself change when the mapping between visual flow and physical movement is manipulated. RDW directly alters this mapping, yet little is known about how gaze behaviour reorganizes during prolonged exposure to RDW gains. Questions such as whether gaze becomes more anticipatory, more uncertain, or more tightly aligned to the intended path under manipulation remain unexplored. Recent VR headsets provide high-frequency binocular eye tracking, enabling precise measurement of gaze behaviour during both natural and redirected walking [16]. This opens up an opportunity to examine not only how gaze predicts locomotion, but how gaze-locomotion coupling itself adapts when visual-motor contingencies change. Our research builds on three observations from the literature. First, gaze improves the estimation of 3D body pose from sparse VR tracking signals, as seen in our own recent work on integrating gaze into full-body pose estimation. Second, gaze improves prediction of upcoming path direction across a range of locomotion tasks. Third, several RDW systems adjust redirection gains based on predicted user motion, but these approaches rely on coarse gaze metrics or limited behavioral models. What is missing is an understanding of how gaze changes in response to RDW manipulation and whether these gaze changes can themselves improve locomotion prediction and RDW control. The overarching goal of our research is therefore to investigate how gaze-locomotion coupling reorganizes during both short- and long-term exposure to RDW, and to determine whether gaze remains a reliable indicator of walking direction as users adapt to altered visual-motor contingencies.

2 COMPLETED AND ONGOING WORK

2.1 Gaze-Augmented 3D Pose Estimation

Previous works shows the potential of Pose estimation using minimum amount of trackers [4, 5, 3, 11]. More studies examined whether binocular gaze can enhance 3D full-body pose estimation when only an HMD and two hand controllers are available. To study this, we collected a synchronized dataset of head, hand,

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and body motion at 120 Hz together with binocular gaze vectors recorded at 50 Hz. Building on this dataset, we proposed *Pose-FusionNet* [2], a multi-stream architecture that fuses gaze direction with head and controller motion to predict full-body joint rotations. We found that including gaze improved the estimation of full-body joints. These findings reinforce the idea that gaze carries advance information about posture and action intention [9, 8]. This work demonstrates that gaze is not merely a perceptual signal but a potent predictor for both body configuration and imminent movement.

2.2 Redirected Walking and Gaze Behaviour

We are currently conducting measurements on a long-term RDW study in which participants complete multiple sessions of curved walking under various curvature gains, extending prior research on perceptual adaptation [6]. Unlike earlier studies that did not measure gaze directly, our dataset includes high-frequency binocular gaze recordings for every walking trial. Our ongoing analyses examine how gaze direction, gaze–head alignment, fixation distance, and gaze stability evolve across redirected and non-redirected walking conditions. Preliminary observations reveal systematic differences between normal and RDW walking. These findings suggest that RDW may induce a form of gaze adaptation that parallels locomotor adaptation—an idea that has not yet been addressed in existing RDW or gaze research.

3 RESEARCH PLAN

Our research will progress along three interconnected directions. First, we aim to characterize how gaze–locomotion coupling changes under RDW at different timescales. Earlier work has shown that locomotion can be inferred from head orientation or motion patterns alone [7, 10], but little is known about whether gaze becomes more stable, more anticipatory, or differently aligned with the intended path as users adapt to RDW. We will analyze how gaze anticipates curves, responds to landmarks, and interacts with changes in the visual–motor relationship under both short- and long-term redirected conditions.

Second, we aim to examine whether the informational role of gaze changes during adaptation. If gaze strategies reorganize through repeated RDW exposure, gaze may no longer reflect walking direction in the same way as in natural locomotion. Understanding these changes is essential for determining whether gaze remains a reliable behavioural signal for interpreting locomotion in manipulated environments.

Finally, we aim to test whether gaze adaptation under RDW reflects a task-specific strategy or a broader reorganization of visuomotor behaviour. To assess this, we will examine gaze–locomotion coupling in additional VR scenarios such as route learning, obstacle navigation [15, 19], and free exploration. These analyses will help establish whether RDW-induced gaze changes generalize beyond the specific redirection task. Collectively, this research will contribute to a deeper understanding of how gaze functions not only as a perceptual measure but also as an adaptive behavioural signal during VR locomotion.

4 DISCUSSION AND OPEN QUESTIONS

We seek feedback from the Consortium on three main aspects. First, we welcome input on analytical methods for examining changes in gaze–locomotion coupling during RDW, including approaches for quantifying adaptation across sessions. Second, we aim to refine our experimental design to better isolate short- versus long-term adaptation effects and to identify which gaze metrics are most sensitive to these changes. Finally, we seek advice on how to frame our findings so that gaze adaptation under RDW can be related to broader theoretical principles of visuomotor learning and ecological behaviour in VR.

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